$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

→ assume $x \in \overline{A \cup B}$,

this means $x \notin A \cup B$

→ this means $x \notin A$ and $x \notin B$

→ this means $\underline{x \in \bar{A} \text{ and } x \in \bar{B}}$

→ by definition of intersection:

$$x \in \bar{A} \cap \bar{B}$$

thus $\overline{A \cup B} = \bar{A} \cap \bar{B}$

$\bar{A} \cap \bar{B} = \overline{A \cup B}$ = Homework.